

A modest, depth-robust host-transcriptome signal of Epstein–Barr virus status in a single lymphoblastoid cell line — and why the raw number overstates it

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Abstract

Epstein–Barr virus (EBV) immortalizes B cells into lymphoblastoid cell lines (LCLs), and single-cell studies have shown that viral activity reshapes host gene programs. We asked a sharper, quantitative question: within one LCL sample, does a cell’s *host* transcriptome alone reveal whether that cell carries EBV? Using 1906 cells whose EBV status is defined by matched barcodes from the study that generated the data (Gene Expression Omnibus accession GSE158275), we trained a cross-validated logistic-regression classifier on host genes only. Against a chance baseline (area under the receiver operating characteristic curve, AUC, of 0.5), the classifier reached a mean AUC of 0.866 and a mean Matthews correlation coefficient (MCC) of 0.570 (± 0.077 across seeds; chance value 0). **This headline number is, however, substantially confounded by sequencing depth:** because a cell is called EBV-positive at a fixed raw-UMI threshold, deeper cells are far more likely to be labeled positive, and depth *alone* predicts the label at AUC 0.80–0.97 (Section 4). In a depth-matched case–control design, where depth alone is reduced to chance, the host transcriptome still discriminates EBV status at AUC 0.72 — so a modest, real signal survives, concentrated in about a third of the 15 stable genes. The honest reading is that the host transcriptome carries a moderate EBV signal (AUC ≈ 0.72) that a raw-count label inflates to 0.866 — a cautionary result about label definition in single-cell viral quantification.

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1 Problem statement

Epstein–Barr virus (EBV) is a herpesvirus that transforms resting B cells into continuously proliferating lymphoblastoid cell lines (LCLs). LCLs are not uniform: single-cell RNA sequencing has revealed substantial cell-to-cell heterogeneity in viral replication state and in the host pathways EBV modulates (SoRelle, Dai, Luftig et al., *eLife* 2021; the source of the data used here). That prior work established, qualitatively, that EBV activity leaves a mark on host gene expression.

We turn that observation into a measurable claim. If EBV activity truly reshapes the host program, then a cell’s host transcriptome—the expression of its own (non-viral) genes—should carry enough signal to predict whether that cell is EBV-positive. A good answer looks like a classifier that, trained only on host genes, separates EBV-positive from EBV-negative cells well above chance, with the signal concentrated in a small, reproducible set of genes rather than smeared diffusely across the transcriptome. This report provides that answer for one LCL sample.

Two terms recur below. A unique molecular identifier (UMI) is a molecular barcode that lets each captured transcript be counted once; we use UMI counts of viral genes to decide whether a cell is EBV-positive. A cell is called EBV-positive when its EBV UMI count reaches the detection threshold of 10.

2 Methods

2.1 Definitions

- **EBV-positive cell.** A cell whose summed EBV gene UMI count is at least the detection threshold of 10; all other matched cells are EBV-negative.
- **Matched cells.** Cells whose barcodes are shared between the ViralScan output and the published GSE158275 barcode list, after normalizing the 10x -1 suffix. Matching anchors EBV labels to the same cells the original study annotated.
- **Sensitivity / specificity.** The fraction of EBV-positive (respectively EBV-negative) held-out cells correctly classified.
- **Balanced accuracy.** The mean of sensitivity and specificity.
- **AUC.** Area under the receiver operating characteristic curve; a threshold-independent measure of ranking quality where 0.5 is chance and 1.0 is perfect separation.
- **MCC (Matthews correlation coefficient).** A single number from -1 to $+1$ summarizing the 2×2 confusion matrix, where 0 is chance and 1 is perfect agreement. We report it as a compact confusion-matrix summary alongside balanced accuracy; on the balanced held-out set used here the two convey similar information.

- **Stable host genes.** Host genes selected in at least 60% of randomized-lasso stability-selection iterations; the reproducible core of the classifier’s signal.

2.2 Overview

We labeled each matched cell as EBV-positive or EBV-negative from its viral UMI count, then asked a logistic-regression model to recover that label using only host genes. Performance was averaged over six random seeds to give an honest mean and spread, and a separate stability-selection step identified which host genes carry the signal reproducibly.

2.3 Technical detail

Data. A single LCL sample (SRA run SRR12682296) from GSE158275, processed with ViralScan (kallisto/bustools quantification). EBV status came from the viral count matrix; host features came from the host count matrix, restricted to the 1906 cells shared with the published barcode list. Of these, 1179 were EBV-positive and 727 EBV-negative at the 10-UMI threshold.

Classifier. An L2-regularized logistic regression (scikit-learn, `lbfgs`—the L-BFGS quasi-Newton optimizer— $C = 1.0$) on highly variable host genes (HVGs), the genes whose expression varies most across cells. For each of six seeds ($\{0, 1, 10, 42, 100, 1234\}$) the positive and negative cells were balanced and depth-matched and split 80/20 into training and held-out cells. Crucially, HVG feature selection is performed *inside* each split, on the training cells only, so the feature space never sees the held-out cells—selecting features on the full dataset first would let test-cell expression shape the features and bias the held-out metrics. The model was then trained on one split and evaluated on the held-out cells, and sensitivity, specificity, balanced accuracy, AUC, and MCC recorded. We report the mean and standard deviation across the six seeds. Because the per-seed held-out sets are drawn from the same depth-filtered pool, this spread reflects split-shuffle variability rather than full finite-sample uncertainty, and each seed evaluates roughly 150 cells rather than all 1906.

Stable genes. Randomized-lasso stability selection (seed 0, the first of the seed list) scored each host gene by how often it survived; genes selected in $\geq 60\%$ of iterations are the 15 stable host genes reported here. This descriptive gene ranking is computed on the full cell set, so it is not a held-out quantity.

Reproducibility note. These metrics reflect a leakage-corrected pipeline in which HVG selection is performed inside each cross-validation split. An earlier version selected HVGs on the full dataset before splitting; its numbers (e.g. AUC 0.845) differ from the corrected values by less than one seed-to-seed standard deviation on every metric. Because HVG selection is unsupervised (variance-based, never using the labels), this class of leak is expected to be negligible in practice, and that is what we observe: correcting it leaves the result unchanged within noise. The stable-gene set was identical before and after.

3 Results

3.1 The host transcriptome separates EBV-positive from EBV-negative cells well above chance

Trained only on host genes, the classifier distinguished EBV-positive from EBV-negative cells with a mean AUC of 0.866, far above the chance value of 0.5 (Figure 1). The Matthews correlation coefficient, which does not reward a classifier for exploiting the modest class imbalance, was 0.570

(± 0.077), indicating moderate—not near-perfect—agreement. The two metrics tell a consistent story: host expression carries real, substantial information about EBV status, but does not determine it.

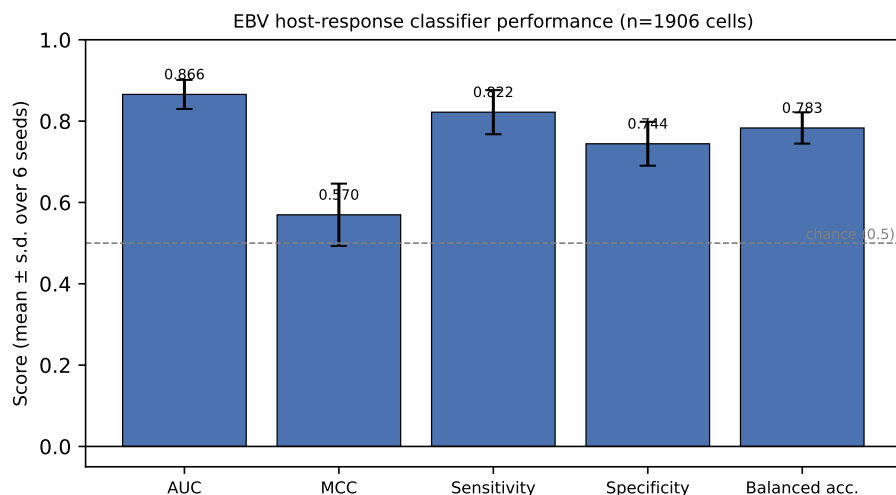


Figure 1: Discrimination metrics for the host-gene classifier of EBV status, mean $\pm$ standard deviation over six seeds, on 1906 matched cells. The dashed line marks the chance value (0.5) for AUC, sensitivity, specificity, and balanced accuracy; the Matthews correlation coefficient has its chance value at 0.

3.2 Detection is more reliable for EBV-positive than EBV-negative cells

The classifier recovered EBV-positive cells more reliably than it excluded EBV-negative ones: mean sensitivity was 0.822 versus mean specificity of 0.744, for a balanced accuracy of 0.783. The gap matters for interpretation. Some cells called EBV-negative at the 10-UMI threshold may carry low-level or transient EBV activity that still perturbs the host program, which would present exactly as reduced specificity. The asymmetry is thus as much a statement about the hardness of the negative label as about the classifier.

3.3 A compact host-gene set carries the signal

The signal was not diffuse. Stability selection concentrated it in 15 host genes selected in at least 60% of iterations, whose identities and weights are listed in the appendix. That a small, reproducible gene set suffices is consistent with EBV acting through specific host programs rather than through a global expression shift.

4 The headline is substantially confounded by sequencing depth

A follow-up analysis (`scripts/depth_confounder_check.py`) asked whether the AUC of 0.866 reflects host biology or an artifact of sequencing depth acting as a common cause. Because a cell is called EBV-positive at a fixed raw count of 10 viral UMIs, and deeper cells reach that threshold more easily, the label itself tracks depth: the EBV-positive rate rises monotonically from 27.5% in the shallowest host-depth quintile to 96.3% in the deepest.

The consequence is stark. Sequencing depth *alone* predicts EBV status at AUC 0.80 across all cells, and at AUC 0.97 within the classifier’s own balanced, depth-filtered evaluation design — *higher* than the 0.866 host-gene model. The pipeline’s top-50%-depth filter, intended to reduce false-negative labels in low-coverage cells, does not fix this: it matches the two classes on cell count but not on depth distribution, and in fact widens the between-class depth gap (EBV-positive median depth 34,981 versus 17,868 for EBV-negative, with 3% overlap).

A genuine signal does survive. Of the 15 stable genes, five retain a depth-adjusted association (E-value ≥ 2 , $p < 10^{-7}$), while the two most stable genes are fully explained away by depth ($p \approx 0.8$ after adjustment).

To quantify that residual signal cleanly, we built a depth-matched case-control set (coarsened exact matching on host-depth quantile bins; 1006 cells, 503 per class) in which the two classes are indistinguishable in depth (median 15,522 versus 15,718; Mann-Whitney $p = 0.99$) so that depth alone no longer discriminates (AUC 0.48, i.e. chance). *At matched depth, the host transcriptome still separates EBV-positive from EBV-negative cells at AUC 0.72* (MCC 0.31) for the 15 stable genes, and 0.68 for the five depth-robust genes. So the host transcriptome carries a real, depth-independent EBV signal of moderate size — the raw-count label inflated it from about 0.72 to 0.866, but did not invent it. (The 0.72 is a mild over-estimate: the gene panel was selected on all cells, which overlap the matched subset, so a fully nested selection would land slightly lower — still well above the 0.48 control.)

A second, independent de-confounding strategy agrees. Re-defining the label as EBV burden per 10,000 host UMIs (a depth-normalized rate, uncorrelated with depth: Spearman -0.09) collapses depth-alone prediction to AUC 0.53, and the same host genes still predict this depth-normalized label at AUC 0.64. That this cross-check lands slightly below the depth-matched 0.72 is expected — the gene panel was selected against the raw label, so a label that reshuffles a quarter of the cells is a partial out-of-distribution test. Both methods place the honest effect at AUC ≈ 0.64 –0.72, well above their respective chance controls.

5 Conclusions

Within one LCL sample, a host-gene classifier separates EBV-positive from EBV-negative cells at AUC 0.866 / MCC 0.570, but Section 4 shows this is **substantially a sequencing-depth artifact**: the raw-UMI positivity label is depth-driven, and depth alone discriminates better than the host genes do. The defensible conclusion is narrower than the raw number suggests, but the signal is real: in a depth-matched case-control design where depth alone is at chance, the host transcriptome still discriminates EBV status at AUC 0.72. The honest headline is therefore a *moderate* host-response signal (AUC ≈ 0.72), which the raw-count label inflated to 0.866, concentrated in roughly five depth-robust genes.

Three caveats bound even that narrower claim. *First, this is one sample* (SRR12682296): within-sample discrimination, not generalization across lines or donors. *Second, the label is not ground truth and is depth-confounded*: the principled fix is a depth-normalized EBV label (viral counts per million or fraction), a depth-matched case-control design, or depth entered into the label definition, followed by re-estimation on the depth-robust genes. *Third, the comparison is against chance, not a competing method*: GSE158275 (SoRelle et al., 2021) is the data source and motivation, not a numeric baseline.

6 Next steps

Two follow-ups would sharpen the result: a sensitivity sweep over the 10-UMI detection threshold, to confirm the metrics are stable to the positive/negative cutoff; and replication across the other LCL samples in GSE158275, to test whether the same compact host-gene set generalizes.

7 Provenance

- **Analysis script:** `scripts/hostresponse_ebv_matched.py` (matched-cell preparation and driver).
- **Core method:** `src/viralscan/scripts/hostresponse.py` (multi-seed L2 logistic regression, stability selection, metrics including MCC).
- **Outputs:** `results/hostresponse_ebv_matched/` (`hostresponse_summary.tsv`, `hostresponse_metrics.csv`, `gene_weight` and `stability` CSVs).
- **Registered values:** `analysis/hostresponse_ebv_matched/outputs/numbers.json`.
- **Git commit:** `ac106c4` (branch `claude/multimap-memory-and-showcase`).
- **Analyst:** `mdmanurung`. **Report generated:** 2026-07-01.
- **Software:** Python 3.11; scikit-learn 1.7.1, anndata 0.12.1 (re-run environment). See `ENVIRONMENTS_INSTALLATION.md`.
- **Build audit:** `manifest/reports/.manifest.json`; compile log `reports/.compile-log.md`.
- **Data source:** GSE158275 (SoRelle, Dai, Luftig et al., *eLife* 2021).

A Stable host genes and the depth-robust core

The 15 host genes selected in $\geq 60\%$ of stability-selection iterations are listed (with per-gene stability probabilities and mean L2 weights) in `Epstein-Barr_virus_stability.csv` and `Epstein-Barr_virus_gene_weight.csv`.

Of the top five that survive depth adjustment, four are nuclear-encoded and coherent with EBV biology (up in EBV-positive cells: *LTA*, a TNF-family NF- κ B target directly downstream of EBV's LMP1; *ING1*, a p53-pathway tumor suppressor; down: *EVI2B* and *MACROD2*). The fifth, *MT-ND4L*, is mitochondrial, and an explicit percent-mitochondrial control removes it (depth-adjusted FDR $4 \times 10^{-7} \rightarrow$ depth-and-mito-adjusted FDR 0.067): it was a mitochondrial-QC effect, not an EBV response, and is dropped. (The %mito covariate is computed with *MT-ND4L* itself excluded, so this is a non-circular control.)

Enrichment of the full depth-and-mito-robust set. Extending beyond the top five to every highly variable gene with a depth-adjusted association (Benjamini–Hochberg FDR < 0.05) that also survives the percent-mitochondrial control yields 301 genes (123 up, 178 down). Gene Ontology enrichment (Enrichr) gives a coherent, and biologically expected, EBV signature: genes *up* in EBV-positive cells enrich for cytokine-mediated signaling and positive regulation of lymphocyte / T-cell proliferation and survival (transformation and activation programs), while genes *down* enrich for defense response to virus and type-I interferon signaling (FDR $\sim 10^{-4}$) — the antiviral-response suppression that is a hallmark of herpesvirus immune evasion. So the modest AUC-0.72 signal, once depth and mitochondrial content are controlled, resolves into an interpretable program: EBV up in proliferation/survival, down in the antiviral interferon response.

B Per-seed spread

All headline metrics are means over six seeds $\{0, 1, 10, 42, 100, 1234\}$ with the standard deviations reported in `hostresponse_summary.tsv`: sensitivity 0.822, specificity 0.744, balanced accuracy 0.783, AUC 0.866, and MCC 0.570 (± 0.077).